

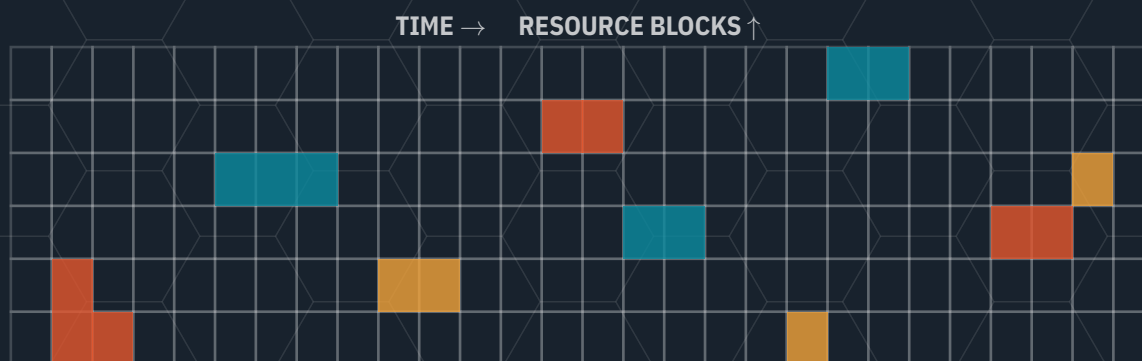
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 7

LTE Scheduling

زمان‌بندی LTE



Lecture Notes

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1 Why Scheduling Matters

In LTE, the radio channel is a **shared resource**. Multiple users compete for limited time-frequency resources on the same carrier. The scheduler is the central intelligence at the eNodeB that decides:

- **Who** gets to transmit/receive
- **When** (which subframe)
- **Where** (which resource blocks)
- **How** (which modulation and coding scheme)

Without scheduling, users would collide, resources would be wasted, and QoS guarantees would be impossible. The scheduler must balance:

- **Throughput maximization** — use resources efficiently
- **Fairness** — don't starve cell-edge users
- **Latency** — meet delay budgets for real-time services
- **QoS** — honor GBR (Guaranteed Bit Rate) bearers

2 The LTE Resource Grid

The resource grid is the fundamental 2D structure that organizes all LTE transmissions in time and frequency.

Time Domain Hierarchy

Structure	Duration	Composition
Radio Frame	10 ms	10 subframes
Subframe	1 ms	2 slots
Slot	0.5 ms	7 OFDM symbols (normal CP)
OFDM Symbol	$\sim 71.4 \mu\text{s}$	1 symbol period + cyclic prefix

Frequency Domain Hierarchy

Structure	Bandwidth	Composition
Subcarrier	15 kHz	Smallest frequency unit
Resource Block (RB)	180 kHz	12 subcarriers
System Bandwidth	1.4–20 MHz	6–100 RBs

Key Definitions

- **Resource Element (RE):** 1 subcarrier $\times$ 1 OFDM symbol — the smallest allocatable unit
- **Resource Block (RB):** 12 subcarriers $\times$ 7 symbols = **84 REs** (one slot)
- **Resource Block Pair:** 12 subcarriers $\times$ 14 symbols = **168 REs** (one subframe / TTI)

Resource Grid Visualization

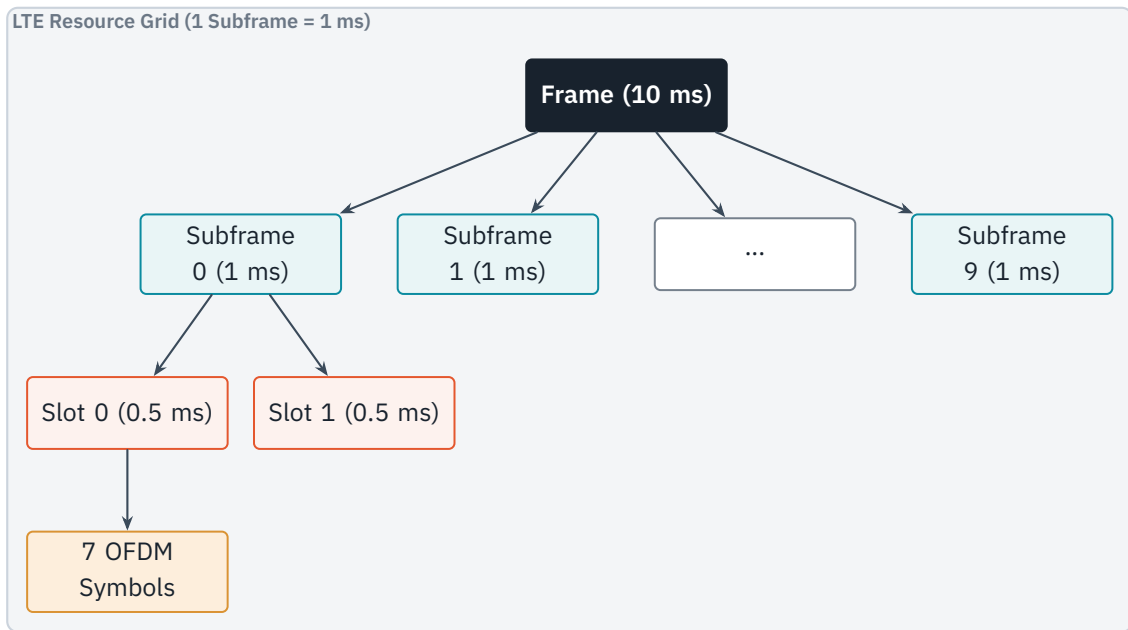


Figure 7.1 LTE time structure: frame, subframe, slot and OFDM symbols

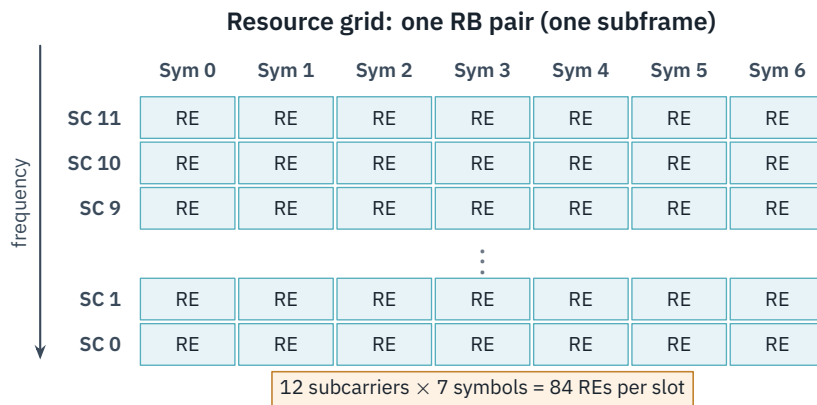


Figure 7.2 Resource elements of one resource block (12 subcarriers × 7 symbols)

3 Scheduling Granularity

Parameter	Value
Time granularity	1 subframe = 1 ms (Transmission Time Interval, TTI)
Frequency granularity	1 Resource Block = 180 kHz (12 subcarriers)
Minimum allocation	1 RB pair per TTI per user
Maximum allocation	All RBs in a TTI to one user
Scheduling decision rate	1000 decisions/second

The scheduler makes a fresh decision every TTI (1 ms), assigning RBs to UEs based on channel conditions, buffer status, QoS requirements, and fairness metrics.

4 Downlink Scheduling

Overview

The eNodeB scheduler controls all downlink resource assignments. It informs UEs of their allocations via **DCI (Downlink Control Information)** messages carried on the **PDCCH (Physical Downlink Control Channel)**.

DCI Contents

A DCI message tells the UE:

- Which RBs are allocated (resource assignment)
- Which MCS to use for decoding
- HARQ process number and redundancy version
- Power control commands
- MIMO-related parameters (precoding, layers)

Resource Allocation Types

Type	Method	Flexibility	Overhead
Type 0	Bitmap of RB Groups (RBGs)	Distributed, non-contiguous	Medium
Type 1	Subset + shift + bitmap	Distributed within subset	Higher
Type 2	Start RB + length (RIV)	Contiguous only	Lowest

RBG size depends on system bandwidth:

System BW (RBs)	RBG Size (RBs)
6	1
15	2
25	2
50	3
75	4
100	4

MCS Selection

The scheduler selects MCS based on:

1. UE reports **CQI** (Channel Quality Indicator)
2. Scheduler maps CQI → appropriate MCS index
3. Applies outer-loop adjustments based on ACK/NACK history
4. Encodes DCI with selected MCS for the UE to decode PDSCH

5 Uplink Scheduling

Key Differences from Downlink

Unlike downlink, uplink scheduling must account for:

- **SC-FDMA constraint:** UEs must be allocated **contiguous** RBs (single-carrier property)
- **UE power limitations:** Distance from eNodeB affects achievable rate
- **Buffer awareness:** eNodeB doesn't directly know UE buffer status

Scheduling Grant

The eNodeB sends an **uplink grant** via DCI (Format 0/4) on PDCCH, containing:

- RB allocation (contiguous)
- MCS index
- Power control command
- HARQ information

Buffer Status Reports (BSR)

UEs inform the eNodeB about queued data via BSR:

BSR Type	Trigger
Regular BSR	New data arrives in a logical channel group with higher priority than currently reported
Periodic BSR	Timer-based periodic reporting
Padding BSR	Sent when UL grant has spare capacity

BSR reports data volume per **Logical Channel Group (LCG)**, allowing priority-aware scheduling.

Power Headroom Report (PHR)

PHR tells the eNodeB how much **additional transmit power** the UE has available:

$$PH = P_{MAX} - P_{PUSCH}$$

- If $PH > 0$: UE can support higher MCS or more RBs
- If $PH \leq 0$: UE is power-limited, scheduler should reduce allocation

SC-FDMA Contiguous Allocation



Figure 7.3 SC-FDMA uplink allocations must be contiguous in frequency

6 Channel Quality Indicator (CQI)

What CQI Represents

CQI is a **4-bit index** (1–15) reported by the UE indicating the highest MCS that can be received with a transport block error rate (BLER) $\leq 10\%$.

CQI = 0 means “out of range” (channel too poor for any transmission).

CQI Table (3GPP TS 36.213, Table 7.2.3-1)

CQI Index	Modulation	Code Rate $\times 1024$	Spectral Efficiency (bps/Hz)
1	QPSK	78	0.1523
2	QPSK	120	0.2344
3	QPSK	193	0.3770
4	QPSK	308	0.6016
5	QPSK	449	0.8770
6	QPSK	602	1.1758
7	16QAM	378	1.4766
8	16QAM	490	1.9141
9	16QAM	616	2.4063
10	64QAM	466	2.7305
11	64QAM	567	3.3223
12	64QAM	666	3.9023
13	64QAM	772	4.5234
14	64QAM	873	5.1152
15	64QAM	948	5.5547

Wideband vs Subband CQI

Reporting Mode	Description	Use Case
Wideband CQI	Single CQI for entire bandwidth	Low overhead, coarse adaptation
Subband CQI	CQI per group of RBs	Frequency-selective scheduling, higher gain
UE-selected subband	UE reports best-M subbands	Compromise between overhead and gain

Subband CQI enables **frequency-selective scheduling**, where the scheduler assigns each UE the RBs where their channel is best.

7 Link Adaptation

Adaptive Modulation and Coding (AMC)

AMC dynamically adjusts the modulation order and code rate to match instantaneous channel conditions:

- **Good channel** → Higher-order modulation + higher code rate → More bits/RE
- **Poor channel** → Lower-order modulation + lower code rate → Fewer bits/RE but robust

Modulation Schemes

Modulation	Bits per Symbol	Required SNR (approx.)	Usage
QPSK	2	< 10 dB	Cell-edge users
16QAM	4	10–17 dB	Mid-range users
64QAM	6	17–25 dB	Near-cell users
256QAM	8	> 25 dB	Cat-11+ (Rel-12), best conditions

MCS Index Table (3GPP TS 36.213, Table 7.1.7.1-1)

MCS Index	Modulation Order	TBS Index	Notes
0	2 (QPSK)	0	Lowest rate
1	2 (QPSK)	1	
2	2 (QPSK)	2	
3	2 (QPSK)	3	
4	2 (QPSK)	4	
5	2 (QPSK)	5	
6	2 (QPSK)	6	
7	2 (QPSK)	7	
8	2 (QPSK)	8	
9	2 (QPSK)	9	
10	4 (16QAM)	9	Modulation transition
11	4 (16QAM)	10	
12	4 (16QAM)	11	
13	4 (16QAM)	12	
14	4 (16QAM)	13	
15	4 (16QAM)	14	
16	4 (16QAM)	15	
17	6 (64QAM)	15	Modulation transition
18	6 (64QAM)	16	

MCS Index	Modulation Order	TBS Index	Notes
19	6 (64QAM)	17	
20	6 (64QAM)	18	
21	6 (64QAM)	19	
22	6 (64QAM)	20	
23	6 (64QAM)	21	
24	6 (64QAM)	22	
25	6 (64QAM)	23	
26	6 (64QAM)	24	
27	6 (64QAM)	25	
28	6 (64QAM)	26	Highest rate
29	2 (QPSK)	—	Reserved (retx)
30	4 (16QAM)	—	Reserved (retx)
31	6 (64QAM)	—	Reserved (retx)

Outer Loop Link Adaptation (OLLA)

The inner loop (CQI → MCS) can be inaccurate due to:

- CQI reporting delay
- UE measurement errors
- Interference variability

OLLA adjusts the effective CQI with an offset:

$$MCS_{effective} = f(CQI + \Delta_{OLLA})$$

The offset is updated based on HARQ feedback:

- **ACK received:** $\Delta_{OLLA} += \text{step_up}$ (e.g., +0.1 dB) → try higher MCS
- **NACK received:** $\Delta_{OLLA} -= \text{step_down}$ (e.g., -1.0 dB) → fall back to lower MCS

Target: maintain **10% BLER** at first transmission (ratio step_down/step_up ≈ 9:1).

8 HARQ (Hybrid Automatic Repeat Request)

Overview

HARQ combines **FEC (Forward Error Correction)** with **ARQ (retransmission)** for reliable delivery with low latency.

LTE HARQ Parameters

Parameter	Downlink	Uplink
Type	Asynchronous	Synchronous
# Processes	8	8
Timing	Flexible retx subframe	Fixed: retx at $n+8$
Feedback	ACK/NACK on PUCCH/PUSCH	PHICH (ACK/NACK)
RTT	8 ms	8 ms

Stop-and-Wait with 8 Processes

Each process handles one transport block at a time (stop-and-wait). With **8 parallel processes**, the pipeline stays full:

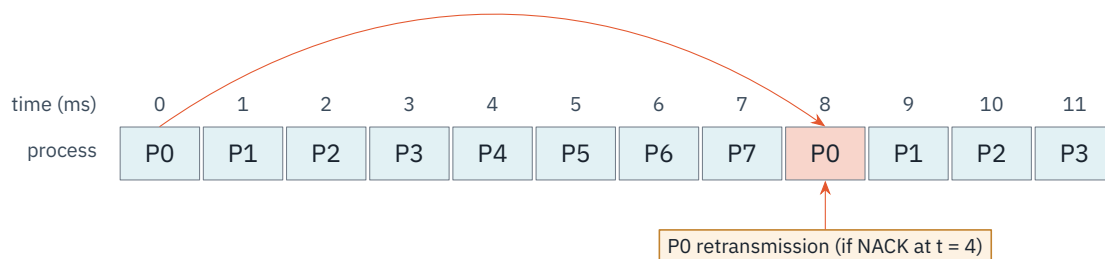


Figure 7.4 Eight parallel stop-and-wait HARQ processes

RTT = 8 ms: UE receives at time t → decodes → sends ACK/NACK at $t+4$ → eNB receives → retransmits at $t+8$.

Combining Methods

Method	Description	Advantage
Chase Combining (CC)	Retransmit identical bits, soft-combine at receiver	Simple, ~3 dB gain
Incremental Redundancy (IR)	Retransmit different redundancy version (new parity bits)	Higher coding gain, better performance

LTE uses **incremental redundancy** by default with 4 redundancy versions (RV 0, 2, 3, 1).

9 Scheduling Algorithms

Round Robin (RR)

Principle: Allocate equal resources to all users in turn.

$$\text{User}_{\text{selected}}(t) = (t \bmod N_{\text{users}})$$

Pros	Cons
Simple to implement	Ignores channel conditions
Perfectly fair in resources	Unfair in throughput (edge users get less)
No feedback required	Poor spectral efficiency
Low computational cost	No QoS differentiation

Proportional Fair (PF)

Principle: Maximize the ratio of instantaneous rate to average throughput.

$$\text{User}_{\text{selected}} = \arg \max_k \frac{R_k^{\text{inst}}(t)}{\bar{R}_k(t)}$$

Where:

- $R_k^{\text{inst}}(t)$ = instantaneous achievable rate for user k (from CQI)
- $\bar{R}_k(t)$ = exponential moving average of user k's throughput

Average update:

$$\bar{R}_k(t+1) = (1 - \frac{1}{t_c})\bar{R}_k(t) + \frac{1}{t_c} \cdot r_k(t)$$

where t_c is the averaging window (typically 100–1000 TTIs).

Pros	Cons
Exploits multi-user diversity	More complex
Good throughput-fairness balance	Requires CQI feedback
Schedules users on their peaks	Tuning t_c affects behavior
Maximizes sum log(throughput)	Not optimal for strict delay

■ Maximum C/I (Max Throughput)

Principle: Always schedule the user with the best channel.

$$\text{User}_{\text{selected}} = \arg \max_k R_k^{\text{inst}}(t)$$

Pros	Cons
Maximizes cell throughput	Starves cell-edge users
Best spectral efficiency	Extremely unfair
Simple decision	Violates QoS for weak users

■ QoS-Aware Scheduling

Principle: Guarantee minimum rates for GBR bearers, prioritize by QCI.

Steps:

1. **Satisfy GBR bearers first** — allocate minimum required RBs
2. **Priority ordering** — rank by QCI priority level
3. **Remaining resources** — distribute via PF among non-GBR bearers
4. **Delay budget check** — boost priority for packets approaching deadline

10 Engineering Trade-offs

The Scheduling Trilemma

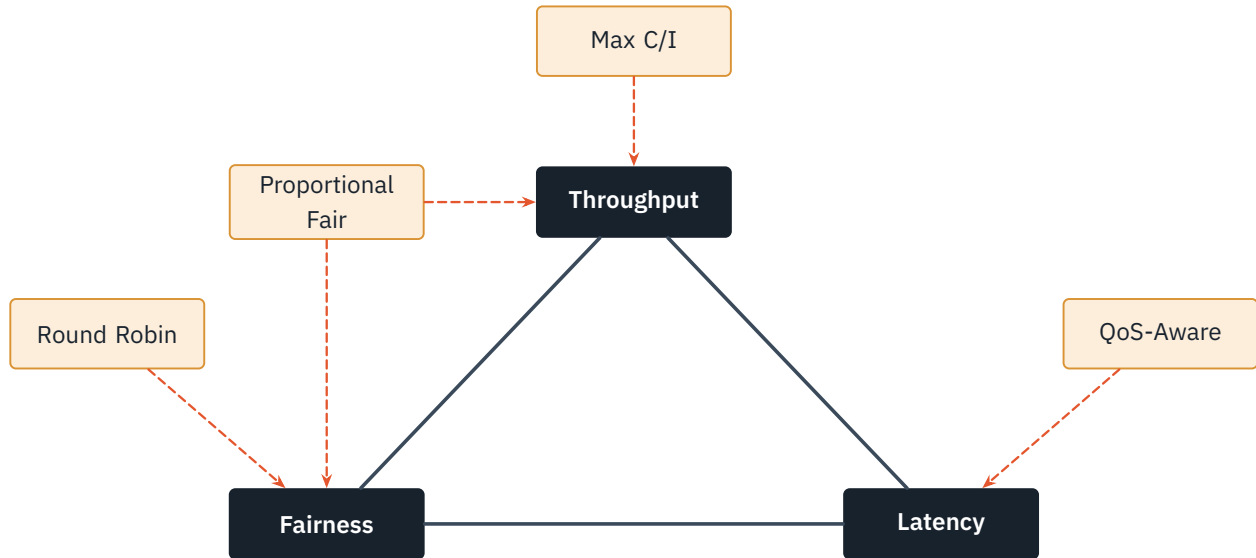


Figure 7.5 The scheduling trilemma and where each algorithm sits

Algorithm	Throughput	Fairness	Latency	Complexity
Round Robin	Low	High (resource)	Medium	Very Low
Proportional Fair	High	Medium	Medium	Medium
Max C/I	Highest	Very Low	Low (for best users)	Low
QoS-Aware	Medium-High	Configurable	Guaranteed	High

Design Considerations

Factor	Impact on Scheduling
User density	More users → more multi-user diversity → PF gains increase
Traffic type	VoLTE needs strict latency; FTP tolerates delay
Cell size	Large cells → big CQI spread → fairness more critical
Frequency band	Higher bands → more path loss variation → more scheduling gain
MIMO mode	MU-MIMO allows co-scheduling users on same RBs

11 Numerical Example

System Parameters

- **Bandwidth:** 10 MHz → **50 RBs** available per TTI
- **Users:** 5 UEs with different channel conditions
- **Overhead:** Assume 10 data symbols per subframe (after control/reference signals)

User CQI and Spectral Efficiency

User	CQI	Modulation	Spectral Efficiency (bps/Hz)	Bits per RB per TTI
UE1	4	QPSK	0.6016	$0.6016 \times 180\text{kHz} \times 1\text{ms} = 108$
UE2	7	16QAM	1.4766	$1.4766 \times 180\text{kHz} \times 1\text{ms} = 266$
UE3	10	64QAM	2.7305	$2.7305 \times 180\text{kHz} \times 1\text{ms} = 491$
UE4	13	64QAM	4.5234	$4.5234 \times 180\text{kHz} \times 1\text{ms} = 814$
UE5	15	64QAM	5.5547	$5.5547 \times 180\text{kHz} \times 1\text{ms} = 1000$

Note: Bits per RB per TTI $\approx$ Spectral Efficiency $\times$ 12 subcarriers $\times$ 15 kHz spacing $\times$ 1 ms $\approx$ SE $\times$ 180 (simplified; actual TBS lookup gives discrete values).

Round Robin Allocation

Each user gets equal RBs: $50 \text{ RBs} \div 5 \text{ users} = 10 \text{ RBs/user}$

User	RBs	Throughput per TTI	Throughput (Mbps)
UE1	10	$10 \times 108 = 1,080 \text{ bits}$	1.08
UE2	10	$10 \times 266 = 2,660 \text{ bits}$	2.66
UE3	10	$10 \times 491 = 4,910 \text{ bits}$	4.91
UE4	10	$10 \times 814 = 8,140 \text{ bits}$	8.14
UE5	10	$10 \times 1,000 = 10,000 \text{ bits}$	10.00
Total	50	26,790 bits	26.79

Jain's Fairness Index (resource): 1.0 (perfectly fair in RBs)

Jain's Fairness Index (throughput): ~0.69

Maximum C/I Allocation

All 50 RBs go to UE5 (best channel):

User	RBs	Throughput (Mbps)
UE1	0	0
UE2	0	0
UE3	0	0
UE4	0	0
UE5	50	50.00
Total	50	50.00

Cell throughput: 50 Mbps (maximum possible)

Jain's Fairness Index: 0.20 (worst case for 5 users)

Proportional Fair Allocation

Assuming all users start with equal average throughput, PF initially behaves like Max C/I. After convergence, it balances by scheduling users on their relative peaks.

Steady-state approximation (PF tends toward equal throughput in log domain):

The PF scheduler approximately equalizes $\log(\bar{R}_k)$, leading to a throughput distribution between RR and Max C/I:

User	Approx. RBs	Throughput (Mbps)	Metric ($R_{\text{inst}}/R_{\text{avg}}$)
UE1	15	1.62	Boosted (low R_{avg})
UE2	12	3.19	Moderate
UE3	10	4.91	Baseline
UE4	8	6.51	Slightly reduced
UE5	5	5.00	Reduced (high R_{avg})
Total	50	21.23	—

Note: Exact PF allocation depends on fading dynamics and tc window. In flat-fading (no frequency selectivity), PF gives more resources to weaker users over time. With frequency-selective fading, all users benefit from being scheduled on their best subbands.

Comparison Summary

Metric	Round Robin	Max C/I	Proportional Fair
Cell Throughput (Mbps)	26.79	50.00	~21–35
UE1 Throughput (Mbps)	1.08	0	~1.6
UE5 Throughput (Mbps)	10.00	50.00	~5.0
Fairness (Jain's, resource)	1.0	0.20	~0.75
Spectral Efficiency	Low	Maximum	High
Starvation?	No	Yes (4 users)	No

Key Takeaways

1. **The resource grid** is a time-frequency matrix — scheduling operates at RB granularity every 1 ms TTI
2. **CQI feedback** is the scheduler's eyes — it drives MCS selection and resource allocation
3. **HARQ** provides reliability with 8 parallel processes and 8 ms RTT
4. **Link adaptation (AMC + OLLA)** continuously matches rate to channel conditions
5. **No single algorithm wins** — the choice depends on deployment scenario, traffic mix, and operator priorities
6. **Proportional Fair** is the de-facto standard in commercial networks as it provides a good throughput-fairness compromise
7. **SC-FDMA** constrains uplink to contiguous allocation, limiting scheduling flexibility compared to downlink OFDMA

References

- 3GPP TS 36.213: Physical layer procedures (CQI/MCS tables)
- 3GPP TS 36.321: MAC protocol (BSR, PHR, HARQ)
- 3GPP TS 36.212: Multiplexing and channel coding (TBS tables)
- Sesia, Toufik, Baker: “LTE – The UMTS Long Term Evolution” (Wiley)